

Vandan Agrawal

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Education

University of Florida

Master of Science in Computer Science | GPA: 3.66

December 2025

Amity University

Bachelor of Technology in Computer Science and Engineering

May 2024

Skills

- **Languages:** Python, Java, C++, C#, x64 assembly language, SQL, Apex, C, Golang, Typescript
- **Frameworks and Libraries:** Next.js, Fast API, Tkinter, NumPy, TensorFlow, Scikit-learn, Salesforce LWC
- **Developer Tools:** Git, GitHub, Claude, Google Cloud Platform, Vertex AI, AWS, Jupyter Notebook
- **Specialized Systems:** Unity 3D, Blender, Ghidra, x64dbg, Wireshark, Splunk, VMware
- **Operating Systems:** Windows, Linux (Debian, Kali, Ubuntu), MacOS

Experience

Cybersecurity Analyst Intern, Malware Reverse Engineering Lab

January 2025 – May 2025

- Reverse-engineered over 10 real-world malware samples analyzing direct system calls in x64 assembly to identify execution flow, persistence, C2, and IOCs, enabling rapid incident response against threats.
- Performed 60+ hours of detailed static and dynamic analysis using Ghidra, Wireshark, x64dbg, etc. on a multi-system sandbox to extract features like execution paths and network indicators.
- Delivered technical reports highlighting obfuscation techniques and behavioral signatures, providing standardized documentation designed to assist incident response teams in threat detection and mitigation.
- Dissected a variant of the Akira ransomware as a final project, documenting behavior and IOCs which has impacted over 250 small to medium-sized American organizations since March 2023.
- Authored YARA rules for various malware by documenting behavioral and threat actor signatures, improving endpoint security to detect payload variants and strengthening organizational security postures.

Software Developer Intern, SmartInternz

May 2023 – July 2023

- Customized and virtually deployed over 20 Salesforce objects, workflows, and permission models to enforce secure, role-based access control and streamline business operations using Object manager.
- Engineered asynchronous Apex triggers and batch processes to manage high-concurrency data without exceeding multi-tenant governor limits improving system responsiveness by 15%.
- Automated security compliance monitoring by implementing Shield Platform Encryption and Security Health Check tools, reducing manual data-integrity reviews by 60% for 5000+ sensitive records.
- Orchestrated asynchronous REST integrations and complex Screen Flows that impacted operational efficiency to automate 500+ weekly transactions across 3 cross-functional simulated departments.

Projects

Municipal Research & Analysis Engine - Google Cloud Platform (GCP), Vertex AI (RAG), FastAPI, Next.js

June 2026

- Architected a municipal search engine on GCP Cloud Run with Next.js and FastAPI, serving secure search analytics with distributed rate-limiting, OAuth 2.0 protection, and zero-downtime scaling.
- Engineered an automated AI research pipeline with Vertex AI and Gemini that processes 650+ municipal documents, automating daily updates to keep search results fresh and accurate without manual effort.
- Integrated Google OAuth and Cloud Firestore to create a persistent research notebook that lets a user manually bookmark documents and store past searches for easy retrieval, ensuring no important findings are lost.

AI-Powered Developer Portfolio - Next.js, TypeScript, Gemini, Vercel

May 2026

- Designed and deployed a portfolio end-to-end with AI-assisted development, from architecture to production.
- Developed a data-driven architecture utilizing a single JSON knowledge base to simultaneously drive all UI components and a Gemini AI chatbot, ensuring context-aware responses and synchronized content.
- Shipped a production-grade web app with zero hosting cost, leveraging Vercel's and Google Gemini's API.

GatorGlide Delivery Co. - Logistics Order Management System - Python

April 2024

- Engineered a dual AVL Tree-based system in Python to synchronize order priorities and ETAs in $O(\log n)$.
- Implemented automated state-validation by utilizing $O(1)$ hash-map lookups and conditional logic to compare real-time system timestamps against calculated departure windows for invalid order cancellations.
- Architected an automated validation suite to parse input test data and cross-reference system output against baseline text files to ensure 100% logical accuracy for regression testing cycles.

MEDEX - A Pharmacy Retail Software - Python, MySQL

July 2022

- Developed a full-stack pharmacy sales application using Python and MySQL, implementing secure, role-based access control (RBAC) to restrict feature access based on employee positions.
- Architected robust backend data retention systems for inventory, customers, employees, and billing via mysql-connector, incorporating strict input field validation to mitigate vulnerabilities.
- Engineered secure employee onboarding workflows by designing a "First Day Password" login system, ensuring data integrity and zero-day access security from day one.